

INDR 252 Applied Statistics - Midterm 1 Study Questions

1. Two quality control engineers collected two random samples, each consisting of 25 observations. Based on their samples, Engineer 1 (based on sample 1) claims something is wrong with the production process, while Engineer 2 (based on sample 2) claims that the production process is under control. Since the production manager is conservative, she decides to check the production system and finds that everything works fine. Choose the incorrect statements:

- a) Sample 1 reflects the population perfectly.
- b) Sample 2 reflects the population perfectly.
- c) The two samples lead to different conclusions due to sampling variation.
- d) Combining the two samples may lead to better conclusions.

2. The density of corn syrup is known to be 1.4 gr/cm³. Nine samples are measured from a bottle. Their average is 1.5 gr/cm³ with standard deviation $s = 0.5$ gr/cm³. Which statement best describes this measuring process?

- a) The bias is 0.1 gr/cm³ and the uncertainty is $\frac{1}{6}$ gr/cm³.
- b) The bias is 0.1 gr/cm³ and the uncertainty is $\frac{1}{2}$ gr/cm³.
- c) Neither bias nor uncertainty can be estimated.
- d) The bias cannot be estimated, and the uncertainty is $\frac{1}{6}$ gr/cm³.
- e) The bias cannot be estimated, and the uncertainty is $\frac{1}{2}$ gr/cm³.
- f) The bias is 0.1 gr/cm³, and the uncertainty cannot be estimated.

3. Let X and Y be independent and identically distributed with an unknown mean μ and a known variance σ^2 . Define the two estimators: $\widehat{\theta}_1 = \frac{X+Y}{2}$ and $\widehat{\theta}_2 = \frac{X+2Y}{3}$ for the unknown mean μ . Choose all correct statements.

- a) $\widehat{\theta}_1$ is unbiased but not $\widehat{\theta}_2$.
- b) $\widehat{\theta}_2$ is unbiased but not $\widehat{\theta}_1$.
- c) Both $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are unbiased.
- d) The standard error of $\widehat{\theta}_1$ is smaller than the standard error of $\widehat{\theta}_2$.
- e) The standard error of $\widehat{\theta}_2$ is smaller than the standard error of $\widehat{\theta}_1$.
- f) The standard error of $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are the same.
- g) The MSE of $\widehat{\theta}_1$ is smaller than the MSE of $\widehat{\theta}_2$.
- h) The MSE of $\widehat{\theta}_2$ is smaller than the MSE of $\widehat{\theta}_1$.
- i) The MSE of $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are the same.

4. Let the random sample $\{X_1, X_2, \dots, X_7\}$ have a geometric distribution with parameter p . What is the likelihood of observing the random sample as $\{2, 1, 3, 2, 5, 6, 1\}$ in terms of p ?

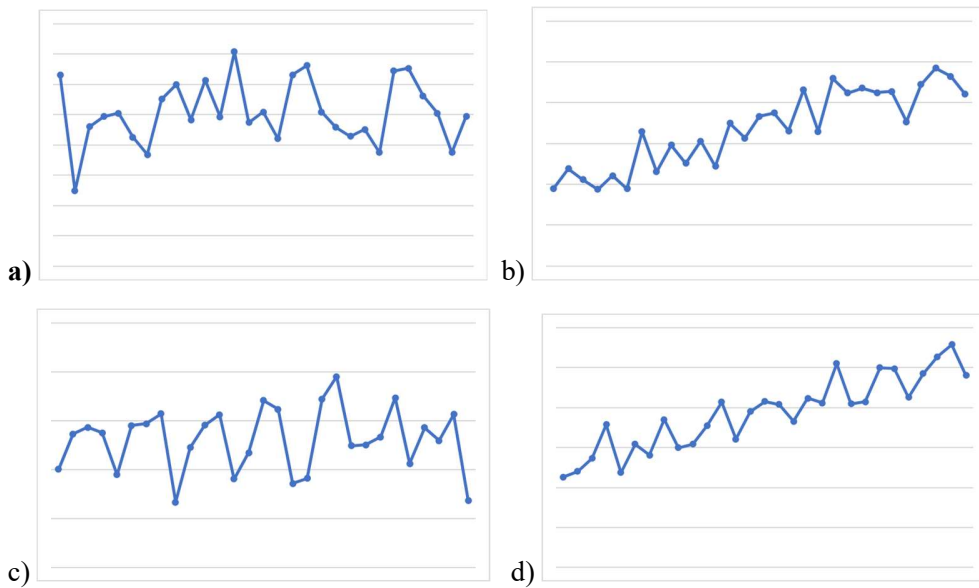
- a) $L(p) = (1-p)^7 p$

- b) $L(p) = (1 - p)^{20}p$
- c) $L(p) = (1 - p)^{20}p^7$
- d) $L(p) = (1 - p)^{13}p^7$
- e) $L(p) = (1 - p)^{20}p^{13}$

5. Let the random sample $\{X_1, X_2, \dots, X_7\}$ have a geometric distribution with parameter p . What is the likelihood estimator of p for the random sample as $\{2, 1, 3, 2, 5, 6, 1\}$ in terms of p ?

- a) $\widehat{p}_{mle} = 1/8$
- b) $\widehat{p}_{mle} = 1/21$
- c) $\widehat{p}_{mle} = 7/27$
- d) $\widehat{p}_{mle} = 7/20$
- e) $\widehat{p}_{mle} = 13/33$

6. Which of the data sets is most likely to have come from a random sample?



7. Here are 10 samples from the height of adult men in centimeters from Sariyer:

[172, 184, 176, 168, 192, 179, 174, 181, 170, 183]

The variance of the population is given as $\sigma^2 = 24.52$. It is also believed that the heights of men in Sariyer come from a normal distribution.

Let $\bar{x}$ denote the sample mean.

Answer the next 3 questions accordingly:

7.1 Which of the following formulas give the $100(1-\alpha)\%$ confidence interval for the population mean μ ?

- a) $(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}})$
- b) $(\bar{x} - z_{\alpha/2} \sigma, \bar{x} + z_{\alpha/2} \sigma)$
- c) $(\bar{x} - z_{1-\alpha} \sigma, \bar{x} + z_{1-\alpha} \sigma)$
- d) $(\bar{x} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}})$

7.2 Which of the following is the 90% confidence interval for the population mean μ ?

- a) (174.83, 180.97)
- b) (168.19, 187.61)
- c) (175.32, 180.48)
- d) (169.76, 186.04)

7.3 Let l, u denote the lower and upper bounds that you have found in question 7.2, respectively. Mark these statements as true or false:

- a) There is a 90% probability that the population mean μ is a value between L and U.
- b) We can be 90% confident that the population mean μ is between L and U.
- c) If we increase the confidence level for the same sample, the confidence interval will shrink.
- d) If we construct 100 different 90% confidence intervals from 100 different random samples of the population using the formula in 7.1, we expect approximately 90 of them to contain the true population mean.
- e) Increasing the sample size will always result in a narrower confidence interval for the population mean, assuming we keep the confidence level the same.
- f) If the standard deviation was not given, we still could have estimated the confidence intervals from the given sample by estimating the standard deviation.

8. A manufacturer produces a certain type of battery. The quality standards require the capacity of these batteries to be in between (137, 143) ampere-hours. An independent consumer-protection agency wishes to test whether the manufacturer complies to the quality standards. For this purpose, they measure the capacity of a random sample of 20 batteries from a recently produced batch. Let p be the proportion of batteries that meet the quality standards.

Stem-and-Leaf Display: Capacity

Leaf Unit = 0.10

1 133 5
2 134 1

3 135 6
 6 136 377
 8 137 34
 (3) 138 028
 9 139 127
 6 140 069
 3 141 18
 1 142
 1 143
 1 144 4

N	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
20	138.47	0.595	2.66	133.50	136.70	138.50	140.45	144.40

Which of the following statements are true?

- a) An unbiased estimator for p is 0.65.
- b) The traditional 95% CI on p is (0.441, 0.859).
- c) The new version 95% CI on p is (0.431, 0.819).
- d) The traditional CI is reliable, since it is based on an unbiased estimator.
- e) The traditional CI is not reliable, since it will cover the true proportion less than 95% of the time.

9. Which assumptions are required for prediction intervals based on a Student's t distribution?

- a) The sample is a random sample.
- b) The sample size is large.
- c) The population standard deviation is known.
- d) The population is normally distributed.

10. Why should methods using a Student's t statistic not be used for a small sample containing an outlier?
 (Has only one answer)

- a) The presence of an outlier in a small sample indicates that the sample does not represent the population.
- b) The Student's t distributions tend not to produce small samples containing an outlier.
- c) Normal distributions tend not to produce small samples containing an outlier.
- d) These methods may be used for a small sample containing an outlier provided the outlier is first discarded.

11. The production manager wants to compare the average weights of popcorns from two production lines. Experience shows that the weights of popcorns can be assumed to be normal. He would like to know whether the true mean of the weights of popcorns from two production lines are different from each other. He obtained the following data:

Sample	n	Sample mean	Sample standard deviation s
Line 1	12	5.2	0.5
Line 2	16	5.61	0.6

Note that the corresponding degrees of freedom is 25.

- a) The two production lines perform similarly with 99% confidence.
- b) The two production lines perform similarly with 95% confidence.
- c) The two production lines perform similarly with 90% confidence.

12. Choose the correct statements:

- a) If $t \sim t_{n-1}$, then the probability density curve for T is bell-shaped and symmetric about 0.
- b) If $t \sim t_{n-1}$ and $Z \sim N(0,1)$, then the probability density curve for t has thicker tails than the curve for Z .
- c) If $X_1, \dots, X_n$ are a small sample from a population with mean μ , then $\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$ has a Student's t distribution with $n - 1$ degrees of freedom.
- d) Methods involving the Student's t statistic should not be used for small samples that contain outliers.

13. When σ is known, why is the z confidence interval $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ better than the t interval $\bar{X} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$?

- a) Approximating the known fixed number σ with the random s adds unnecessary variability.
- b) $\sigma < s$, which tends to make the z interval smaller than the t interval.
- c) $z_{\alpha/2} < t_{n-1, \alpha/2}$, which tends to make the z interval smaller than the t interval.
- d) The sample mean tends to be smaller in the z interval than in the t interval.

14. Assume that you are interested in the mean height (μ) of all men in Turkey, where population standard deviation is 17. You randomly ask 15 men in Sarıyer their heights and obtain the sample mean as 174.94. Which of the following statements are true?

- a) The population mean is 174.94.
- b) The probability that the population mean is in (168, 182) is 90%, if the heights are normal.
- c) The population mean is in (168, 182) with 90% confidence, if the heights are normal.
- d) The sample can be assumed to represent Turkey.

15. A random sample of size 10 is to be drawn from a normal distribution with a mean 4. The Student's t statistic $t = (\bar{X} - 4)/(s/\sqrt{10})$ is to be computed. What is the probability that $t > 1.833$?

- a) 1%
- b) 2%
- c) 5%
- d) 10%

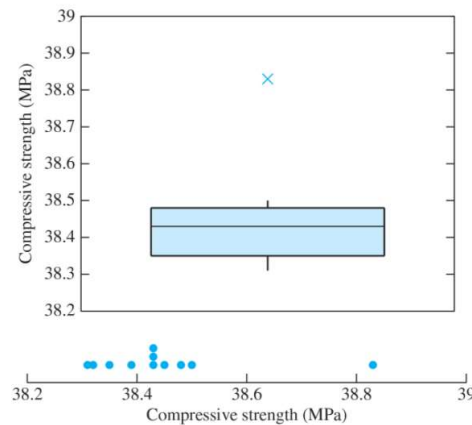
16. To compare two normal populations (X- and Y-populations), 10 observations are collected from each population. Here are the resulting statistics: $\bar{x} = 12.77$, $\bar{y} = 11.77$, $s_x = 4.17$, $s_y = 4.17$. If the differences in each pair are analyzed, we have $\bar{d} = 1$ and $s_d = 0$. Choose all correct statements.

- a) The two-sample method should be used to find a confidence interval around the difference between the two means.
- b) The paired method should be used to find a confidence interval around the difference between the two means.
- c) The two means are different from each other at all confidence levels, since the difference in each pair is exactly equal to 1.
- d) The paired method cannot be used since the corresponding sample standard deviation is 0.

17. A random sample of size 10 will be drawn from a normal distribution with a standard deviation of 4. The Chi-Square statistic $\chi^2 = \frac{9s^2}{\sigma^2}$ is to be computed. Choose all correct statements.

- a) $P(\chi^2 < 3.5) > 0.05$
- b) $P(\chi^2 < 3.5) < 0.05$
- c) $P(\chi^2 > 3.5) > 0.95$
- d) $P(\chi^2 > 3.5) < 0.95$

18. An engineer reads a report that states that a sample of 11 concrete beams has an average compressive strength of 38.45 MPa with standard deviation 0.14 MPa. The box plot and dot plot of the data is given below. Choose all correct statements.



- a) The 95% confidence interval around the population variance is (0.0096, 0.0604).
- b) The 95% confidence interval around the population mean is (36.22, 40.68).
- c) A confidence interval around the population variance cannot be found, since the data has an outlier, indicating that it is not normal.
- d) A confidence interval around the population mean cannot be found, since the data has an outlier, indicating that it is not normal, and the sample size is small.

19. To compare two normal populations (X- and Y-populations), 10 observations are collected from each population. Here are the resulting statistics: $\bar{x} = 10.54$, $\bar{y} = 12.4$, $s_x = 2.1$, $s_y = 1.12$, and $v=13$ (the corresponding degrees of freedom for Student t distribution). If the differences in each pair are analyzed, we have $\bar{d} = 1.86$ and $s_d = 2.86$. The two-sample method gives the following 95% confidence interval around the difference between the two means: (-3.48, -0.23), whereas the paired method gives (-3.9, 0.19). Choose all correct statements.

- a) The two-sample method should be used to find a confidence interval around the difference between the two means if the corresponding assumptions can be verified.
- b) The paired method should be used to find a confidence interval around the difference between the two means if the corresponding assumptions can be verified.
- c) The two means are different from each other at 95% confidence if the corresponding assumptions can be verified.
- d) The two means are not different from each other at 95% confidence.

20. To compare two normal populations (X- and Y-populations), 10 observations are collected from each population. Here are the resulting statistics: $\bar{x} = 10.54$, $\bar{y} = 12.4$, $s_x = 2.1$, $s_y = 1.12$. If the differences in each pair are analyzed, we have $\bar{d} = 1.86$ and $s_d = 2.86$. The two-sample method gives the following 95% confidence interval around the difference between the two means: (-3.48, -0.23), whereas the paired method gives (-3.9, 0.19). Choose all correct statements.

- a) The correlation between the two samples is not high enough.
- b) The two samples can be assumed to be independent.
- c) The two-sample method has a confidence interval with a shorter width because of the degrees of freedom.
- d) The two-sample method has a confidence interval with a shorter width because the associated sample standard deviation is lower.